# Eliza J. Dawson

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## Appointments

**NOAA Climate and Global Change Postdoctoral Fellow** **2024 to present**
*Hosted by Dr. Winnie Chu, Polar Geophysical Simulation Lab, Georgia Tech*

- Principal investigator of research integrating radar sounding observations with numerical ice flow models to improve projections of future ice sheet change and sea level rise.

## Education

**Stanford University, PhD in Geophysics** **2018 to 2024**
*Advised by Dr. Dustin Schroeder*
Thesis title: *Models and Observations of the Antarctic Ice Sheet Thermal State and Implications for Ice Sheet Dynamics*

**University of Washington, Bachelor of Science in Atmospheric Science: Climate** **2014 to 2018**
*With honors, minor in applied mathematics*
Honors thesis advisors: Dr. David Battisti and Dr. Abigail Swan

## Technical Skills

**Scientific computing:** Numerical methods, optimization, inversion, uncertainty analysis, statistical learning
**Signal processing:** Time and frequency domain analysis, attenuation estimation, calibration, quality control
**Programming:** Python, MATLAB; working knowledge of C++
**Computing workflows:** Git, HPC, Slurm, reproducible research pipelines
**Cloud:** Google Cloud Storage (GCS), storage buckets
**Geospatial and data tools:** QGIS, ArcGIS, geopandas, xarray, netCDF
**Sensors and field systems:** Airborne, Ground penetrating, and UAV-borne radar systems, borehole drilling, water sampling, GPS and weather station installation

## Publications

### Journal Articles in Review

1. **Dawson, E. J.,** N. Wolfenbarger, M. R. Feduschak, W. Chu, D. Yang, D. M. Schroeder. (2026) Radar Attenuation through Ice on Earth and Across the Solar System. *Rev. Geophys. (Invited).* In Review.

### Published Journal Articles

1. **Dawson, E. J.,** W. Chu, M. Christoffersen, D. Yang, S. Farris, J. MacGregor. (2026) Ice sheet attenuation from radar sounding in the frequency domain. *J. Glaciol.*, 72, e17, 1–15. doi.org/10.1017/jog.2026.10130.
2. Schroeder, D. M., E. Abrahams, A. L. Broome, W. Chu, R. Culberg, **E. J. Dawson**, E. J. MacKie, D. F. May, M. R. Siegfried, T. O. Teisberg, and S. Zhao. 2026. Next-generation radar bed measurements should be optimized for assimilation or repeat-pass profiling. *Philosophical Transactions of the Royal Society A*, 384, 20240548. doi:10.1098/rsta.2024.0548.
3. Nicola, L., *et al.* (including **E. J. Dawson**). (2025) Where do we want the glaciology community to be in 2073? Equality, diversity, and inclusion challenges and visions from the 2023 Karthaus Summer School. *J.*

*Glaciol.*, 71, e68. doi.org/10.1017/jog.2025.18.

4. **Dawson, E. J.**, D. M. Schroeder, W. Chu, E. Mantelli, and H. Seroussi. (2024) Heterogeneous basal thermal conditions underpinning the Adélie George V Coast, East Antarctica. *Geophys. Res. Lett.*, 51, e2023GL105450. doi:10.1029/2023GL105450.
5. Aitken, A. R. A., L. Li, B. Kulessa, D. M. Schroeder, T. A. Jordan, J. M. Whittaker, S. Anandakrishnan, **E. J. Dawson**, D. A. Wiens, O. Eisen, and M. J. Siegert. (2023) Antarctic sedimentary basins and their influence on ice-sheet dynamics. *Rev. Geophys.*, 61(3). doi:10.1029/2021RG000767.
6. **Dawson, E. J.**, D. M. Schroeder, W. Chu, E. Mantelli, and H. Seroussi. (2022) Ice mass loss sensitivity to the Antarctic ice sheet basal thermal state. *Nature Communications*, 13, 4957. doi:10.1038/s41467-022-32632-2.
7. Bienert, N. L., D. M. Schroeder, S. T. Peters, E. J. MacKie, **E. J. Dawson**, M. R. Siegfried, R. Sanda, and P. Christoffersen. (2022) Post-processing synchronized bistatic radar for long-offset glacier sounding. *IEEE Trans. Geosci. Remote Sens.*, 60, 1 to 17. doi:10.1109/TGRS.2022.3147172.
8. Young, T. J., C. Martín, P. Christoffersen, D. M. Schroeder, S. M. Tulaczyk, and **E. J. Dawson**. (2021) Rapid and accurate polarimetric radar measurements of ice crystal fabric orientation at the WAIS Divide ice core site. *The Cryosphere*, 15(8), 4117 to 4133. doi:10.5194/tc-15-4117-2021.
9. Evans, S., **E. J. Dawson**, and P. Ginoux. (2020) Linear relation between shifting ITCZ and dust hemispheric asymmetry. *Geophys. Res. Lett.*, 47(22). doi:10.1029/2020GL090499.
10. Kim, J. E., M. Lague, S. Pennypacker, **E. J. Dawson**, and A. L. S. Swann. (2020) Evaporative resistance is of equal importance as surface albedo in high-latitude surface temperatures due to cloud feedbacks. *Geophys. Res. Lett.*, 47(4). doi:10.1029/2019GL085663.
11. Donohue, A., **E. J. Dawson**, L. McMurdie, D. S. Battisti, and A. Rhines. (2019) Seasonal asymmetries in the lag between insolation and surface temperature. *J. Climate*, 30, 10117 to 10137. doi:10.1175/JCLI-D-19-0329.1.
12. Potter, S. F., **E. J. Dawson**, and D. M. W. Frierson. (2017) Southern African orography impacts on low clouds and the Atlantic ITCZ in a coupled model. *Geophys. Res. Lett.*, 44. doi:10.1002/2017GL073098.

## Fellowships and Grants

**Living data products for ice sheet models** — **2026**
*Astera Foundation*
Total value of award: $57,750

**Climate and Global Change Postdoctoral Fellowship** — **2024 to 2026**
*National Oceanic and Atmospheric Administration*
Total value of award: $172,000

**Office of Polar Programs Postdoctoral Fellowship** — **Awarded but declined**
*National Science Foundation*
Total value of award: $167,800

**Graduate Research Fellowship Program Recipient** — **2019 to 2022**
*National Science Foundation*

**Atmospheric Sciences Reed Caldwell Scholarship** — **2017 to 2018**
*University of Washington*

**Ernest F. Hollings Undergraduate Scholarship** — **2015 to 2017**
*National Oceanic and Atmospheric Administration*

## Field Experience

**Helheim Glacier, Greenland (supported by the Heising Simons Foundation)** **June to July 2025**

- *Coordinated logistics and scientific operations for radar surveys targeting firn aquifer and bed structure.*
- *Supported complementary geophysical programs, including water sampling, noble gas analysis, and supraglacial lake monitoring.*
- *Maintained and reinstalled GPS stations to track glacier motion.*

**Longyearbyen, Svalbard (supported by The University Centre in Svalbard)** **March 2023**

- *Coordinated ground-based radar field operations to investigate polythermal glacier structure.*
- *Instructed students in ground-based and UAV borne radar surveying techniques.*
- *Supported active seismic surveys along the glacier flowline.*

**Vatnajökull Ice Cap, Iceland (supported by Stanford University)** **Aug to Sept 2022**

- *Tested and validated new ground-based and UAV-borne radar systems in highly attenuating ice.*
- *Coordinated multi system radar field operations and set survey priorities.*

**Thwaites Glacier, Antarctica (supported by NSF ITGC TIME project)** **Dec 2019 to Feb 2020**

- *Led ground-based multi offset radar surveys across the Thwaites Eastern Shear Margin.*
- *Collected polarimetric radar profiles to investigate englacial structure and basal conditions.*

**Store Glacier, Greenland (supported by Cambridge University RESPONDER)** **July to Aug 2019**

- *Conducted ground-based radar surveys in collaboration with project scientists.*
- *Assisted with borehole drilling and instrumentation to support subglacial hydrology studies.*

## Teaching Experience

**Guest Instructor** **2023**
*Arctic Geophysics, University Centre in Svalbard, Norway*

**Teaching Assistant, Stanford University** **2019 to 2020**
*Introduction to the Foundations of Contemporary Geophysics*

- 2020
- 2019

## Student Mentoring

**Neosha Narayanan** **2026 to present**
*PhD Student, Radar geophysics for Greenland outlet glaciers, Georgia Tech*

**Mandala Pham** **2025 to present**
*PhD Student, Radar geophysics and subglacial geology, Georgia Tech*

**Donglai Yang** **2024 to present**
*PhD Student, Ice sheet modeling and AI methods, Georgia Tech*

**Rowan Ray** **2024 to present**
*Undergraduate Student, Ice core and borehole data analysis, Georgia Tech*

**Kiera Tran** **2022 to present**
*PhD Student, Radar characterization of ice shelves, Georgia Tech*

**Chloe Cheng** **2023**
*Undergraduate Student, Ice ocean interactions, Stanford University*

**Lena Schwebs** **2021**
*Undergraduate Student, Radar sounding data analysis, Summer intern*

## Invited Talks

- Helheim "Follow the Water" field season recap and progress update. *Georgia Institute of Technology*. August 2025
- From radar sounding data to the ice sheet basal thermal state. *INSTANT Geothermal Heat Flow Seminar, virtual*. May 2024
- Closing gaps in polar geophysics: how combining models and observations rewrites the story of mass loss from Antarctica. *University of California Los Angeles, Los Angeles, CA*. Apr 2024
- Icy insights by bridging models and observations: Antarctic mass loss sensitivity to the thermal state. *University of California Santa Cruz, Santa Cruz, CA*. Dec 2023
- Icy insights by bridging models and observations: Antarctic mass loss sensitivity to the thermal state. *California Institute of Technology, Pasadena, CA*. Nov 2023
- Is Antarctica vulnerable to basal thawing? Evidence from modeling and observations. *Ludwig Maximilian University, Munich, Germany*. Jun 2023
- Investigating the role of basal thawing in Antarctica. *Georgia Institute of Technology, Atlanta, GA*. Mar 2023
- Investigating basal thawing in Antarctica with ice sheet modeling and ice penetrating radar. *International Glaciological Society Global Seminar, online*. Dec 2022
- Investigating the Antarctic Ice Sheet response to basal thaw. *University of Colorado, Boulder, virtual*. Jun 2020

- The next instability? Modeling basal thermal transitions of ice sheets. *NASA Jet Propulsion Laboratory, CA*. Sept 2019

## Conference Abstracts

- Dawson, E. J., M. Christoffersen, L. Liu, D. Yang, W. Chu. Basal Temperatures from Radar Sounding: New Inferences Across East Antarctica. American Geophysical Union Fall Meeting. New Orleans, LA, Dec 2025
- Dawson, E. J., W. Chu, M. Christoffersen, D. Yang. Advancing Radar Sounding Attenuation Estimates with Frequency-Based Techniques. European Geophysical Union General Assembly. Vienna, Austria, May 2025
- Dawson, E. J., W. Chu, D. Yang, M. Christoffersen. Subsurface insights from new radar sounding attenuation estimates across Antarctica. American Geophysical Union Fall Meeting, Washington DC, Dec 2024
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Evidence for heterogeneous basal thermal conditions along the Adelie George V Coast, East Antarctica. American Geophysical Union Fall Meeting, San Francisco, CA, Dec 2023
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Evidence for heterogeneous basal thermal conditions underpinning the Adelie George V Coast, East Antarctica. SCAR INSTANT Conference. Trieste, Italy, Sept 2023
- Dawson, E. J., E. Wilson. Exploring oceanic heat pathways along the George V Land continental shelf. European Geophysical Union General Assembly. Vienna, Austria, Apr 2023
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Towards the integration of radar subglacial constraints into ice sheet models. FOGGS Conference. Atlanta, GA. Mar 2023
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Deciphering the Basal Conditions of Wilkes Basin, East Antarctica, with Ice-Penetrating Radar and Ice Sheet Modeling. American Geophysical Union Fall Meeting. Chicago, IL. Dec 2022
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Deciphering basal thermal conditions with ice-penetrating radar and ice sheet modeling. West Antarctic Ice Sheet Workshop, Estes Park, CO. Sept 2022
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Utilizing radar sounding to constrain the basal thermal state in parts of East Antarctica. IGS Conference, Reykjavik, Iceland. Aug 2022
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Investigating basal thaw as a driver of mass loss from the Antarctic ice sheet. European Geophysical Union General Assembly, Vienna, Austria, May 2022
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Investigating basal thaw as a driver of mass loss across Antarctica. American Geophysical Union Fall Meeting, New Orleans, LA, Dec 2021
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Investigating basal thaw as a driver of mass loss across Antarctica. West Antarctic Ice Sheet Workshop, Sterling, VA, Oct 2021
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Investigating Basal Thaw as a Mechanism of Ice Mass Loss in Antarctica. American Geophysical Union Fall Meeting, virtual, Dec 2020
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Assessing the potential for basal thermal regime change to accelerate mass loss from the Antarctic Ice Sheet. West Antarctic Ice Sheet Workshop, virtual, Sept 2020
- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Investigating basal thaw as a potential driver of ice flow acceleration in Antarctica. European Geophysical Union General Assembly, virtual, May 2020

- Dawson, E. J., D. M. Schroeder, W. Chu, E. Mantelli, H. L. Seroussi. Vulnerability of the Antarctic Ice Sheet to basal thermal regime change. West Antarctic Ice Sheet Workshop, Julian, CA, Oct 2019
- Dawson, E. J., D. M. Schroeder, A. Miltenberger, W. Chu, H. Seroussi. A Comparison of Radar inferred Temperature Characterization Techniques to Investigate Thermal Regime Changes in Antarctica. International Glaciological Society Symposium, Stanford, CA, Jul 2019

## Summer Schools

Summer school on ice sheets and glaciers in the climate system, Karthaus, Italy — May 2023

International Summer School in Glaciology, McCarthy, Alaska — Cancelled due to COVID 19

## Honors and Awards

**Outstanding Student Presentation Award** — **2023**
*American Geophysical Union, Cryosphere Section*

**Outstanding Student Presentation Award** — **2016**
*American Geophysical Union, Atmospheric Sciences Section*

**Antarctic Service Medal** — **2019 to 2020**
*National Science Foundation*

## Service

### Professional Service

- AGU Section Leadership
- AGU Cryosphere OSPA Coordinator
- Member of SCAR Antarctic Geological Boundary Conditions Steering Committee
- Proposal and manuscript reviewer, NSF proposals; The Cryosphere; Geophysical Research Letters; Journal of Glaciology; IEEE journals; Nature Communications
- Session Chair, Improving Predictability, West Antarctic Ice Sheet Workshop, 2022

### University Service

- Student committee to Hire Tenure Track Geophysics Faculty, Stanford University, 2023 to present
- Co Creator, Stanford Ice Seminar: School wide seminar series for polar researchers, 2023 to present
- Graduate Advisor to the Department of Geophysics Chair, Stanford University, 2023 to present
- Graduate Teaching Liaison, Stanford University, 2020 to 2022
- Organizer of Mentors in Teaching Workshops, Stanford University, 2020 to 2022
- Member of Graduate Student Advisory Committee, Stanford University, 2019 to 2020

### Community Outreach

- Featured scientist in climate change books
- Invited speaker for Peninsula Community College public seminar
- Exhibit organizer for Port Townsend Natural History Museum

- Invited speaker, Northwest Maritime Center

## Professional Affiliations

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| International Association of Cryospheric Scientists, Member | 2022 to present |
| Early career Glaciologists Group, Member | 2022 to present |
| European Geosciences Union, Member | 2020 to present |
| International Glaciological Society, Member | 2020 to present |
| American Geophysical Union, Member | 2015 to present |